Image management in OpenStack is primarily handled by the **Glance** service, which serves as a central repository for the templates used to create virtual machines 1, 2.

Role of the Image Service (Glance)

The **Glance API** facilitates the management of virtual machine images, which are **essentially templates** used for deploying VMs 3, 4. Its core responsibilities include:

- **Central Repository:** Acting as a singular location for storing, retrieving, and managing images 1, 5.
- **Image Discovery:** Providing mechanisms for users and applications to find images based on properties like name, operating system, or architecture 1, 6.
- **Access Control:** Managing the visibility of images, designating them as **public** (accessible to all) or **private** (restricted to specific projects) 6, 7.

Image Storage Methods

Images can be added to the Glance service through two primary methods 6:

1. **Uploading Pre-built Images:** Users can upload existing images obtained from external sources or created beforehand 6.
2. **Capturing Snapshots:** OpenStack allows users to capture a snapshot of a **running virtual machine**, creating a new image that reflects the VM's current state 6.

- **Supported Formats:** Glance supports various disk and container formats, including **RAW, QCOW2, VMDK, and OVA** 1.

Image Retrieval Workflow

When a user or a service needs an image, the workflow follows these steps 1, 6, 7:

1. **Discovery:** Users search the repository using **metadata** such as the operating system version, software packages included, or hardware architecture.
2. **Access Verification:** Glance checks the image's visibility settings to ensure the user has the appropriate permissions (public vs. private).
3. **Selection:** Users can leverage **versioning** to select specific historical versions of an image if updates have been made.

Interaction with the Compute Service (Nova)

Glance and Nova (the Compute service) work closely together to provision resources 2, 3:

- **Template Source:** When **Nova** receives a request to launch a VM, it retrieves the specified image from Glance to use as the foundation for the new instance 1, 3.
- **Driver Coordination:** In a clustered environment, the **Senlin Engine** uses internal **drivers** to interact with the Glance API, allowing it to create, update, or destroy image-backed objects as needed 8, 9.

Advantages of an Image Repository

Maintaining a dedicated image repository provides several benefits 10, 11:

- **Standardized Deployments:** Ensures consistency across the cloud, reducing configuration drift.
- **Improved Efficiency:** Streamlines provisioning by eliminating the need to manually configure each VM from scratch.
- **Version Control and Rollbacks:** Allows administrators to track changes and revert to previous image versions during troubleshooting.
- **Reusability:** Saves time by allowing the same template to be used for multiple VMs.

Exam-Focused Explanation

- **Service Name:** The image service is **Glance** 1.
- **Core Concept:** Images are **templates** for VMs 4.
- **Two Creation Methods: Uploading** (external files) and **Snapshotting** (from running VMs) 6.
- **Metadata & Versioning:** Glance uses metadata (OS, software) for identification and supports versioning for **rollbacks** and consistency 7, 11.
- **Key Integration: Nova** uses Glance images to launch compute instances; **Senlin** manages these images as nodes in a cluster via **drivers** 2, 3, 9.
- **Tools:** Image management is accessible via the **Horizon Dashboard** (GUI), **CLI**, and **REST APIs** 11, 12.